UNIVERSAL INJECTIVITY OF THE MAP TO THE ADIC COMPLETION UNDER A CONDITION ON ASSOCIATED PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The criterion for universal injectivity by testing with finite modules. Algebra, label algebra-theorem-universally-exact-criteria, invoked at proof line 2080. Reduces universal injectivity of the map into the completion to the injectivity of the map after tensoring with an arbitrary finite module over the base.
- A module injects into the product of its localisations at its associated primes, and an intersection of powers of an ideal vanishes under a condition. Algebra, labels algebra-lemma-zero-at-ass-zero (proof line 2091) and algebra-lemma-intersect-powers-ideal-module-zero (proof line 2099). The first lets the vanishing of the kernel be checked at the associated primes, and the second shows that the intersection of the powers of the ideal inside the localisation vanishes because the extended ideal is contained in a prime.

Source reference. Original lemma 00IP: sources/stacks-chow/tags.txt, source lines 687–687. Navigation pointer for \ref{algebra-lemma-intersect-powers-ideal-module-zero} used inside the counted span (line 2099); the argument that consumes it is retained in the span.

Source reference. Original lemma 0311: sources/stacks-chow/tags.txt, source lines 3968–3968. Navigation pointer for \ref{algebra-lemma-zero-at-ass-zero} used inside the counted span (line 2091); the argument that consumes it is retained in the span.

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Source reference. Original theorem 058K: sources/stacks-chow/tags.txt, source lines 6783–6783. Navigation pointer for \ref{algebra-theorem-universally-exact-criteria} used inside the counted span (line 2080); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let $R \to S$ be a ring map, and N an S-module. Assume*

(1) *R is a Noetherian ring,*
(2) *S is a Noetherian ring,*
(3) *N is a finite S-module, and*
(4) *for any finite R-module Q, any $\mathfrak{q} \in Ass_S(Q \otimes_R N)$ satisfies $IS + \mathfrak{q} \neq S$.*

Then the map $N \to N^\wedge$ of N into the I-adic completion of N is universally injective as a map of R-modules.

Proof. We have to show that for any finite R-module Q the map $Q \otimes_R N \to Q \otimes_R N^\wedge$ is injective, see Algebra, Theorem 058K. As there is a canonical map $Q \otimes_R N^\wedge \to (Q \otimes_R N)^\wedge$ it suffices to prove that the canonical map $Q \otimes_R N \to (Q \otimes_R N)^\wedge$ is injective. Hence we may replace N by $Q \otimes_R N$ and it suffices to prove the injectivity for the map $N \to N^\wedge$.

Let $K = \mathrm{Ker}(N \to N^\wedge)$. It suffices to show that $K_\mathfrak{q} = 0$ for $\mathfrak{q} \in \mathrm{Ass}(N)$ as N is a submodule of $\prod_{\mathfrak{q} \in \mathrm{Ass}(N)} N_\mathfrak{q}$, see Algebra, Lemma 0311. Pick $\mathfrak{q} \in \mathrm{Ass}(N)$. By the last assumption we see that there exists a prime $\mathfrak{q}' \supset IS + \mathfrak{q}$. Since $K_\mathfrak{q}$ is a localization of $K_{\mathfrak{q}'}$ it suffices to prove the vanishing of $K_{\mathfrak{q}'}$. Note that $K = \bigcap I^n N$, hence $K_{\mathfrak{q}'} \subset \bigcap I^n N_{\mathfrak{q}'}$. Hence $K_{\mathfrak{q}'} = 0$ by Algebra, Lemma 00IP. □